

Thermodynamics

Vaibhav Chandra

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Prerequisite Knowledge

- Basic mechanics: momentum, force as rate of change of momentum
- Comfort with derivatives and definite integrals (see the Calculus handout)
- The idea of a photon as a quantum of light (see the Electromagnetic Theory & Quantum Physics handout), useful for the energy transfer section but not essential beforehand

1 Thermodynamic Equilibrium

What Temperature Actually Measures

Before building anything else, it's worth pausing on a question that's easy to skip past: what *is* temperature, physically? Not "a number a thermometer reads," but what's actually happening at the level of atoms and molecules when we say something is "hot" or "cold"?

The answer, which took physicists a surprisingly long time to nail down, is that temperature is a measure of the average kinetic energy of random, disordered motion of the particles making up a substance. A hot cup of tea isn't fundamentally different from a cold one because of some separate "heat substance" mixed into it (an old and thoroughly discredited idea called caloric theory) — it's different because its water molecules are, on average, jostling around faster. We'll make this precise in Section 2, but for now, hold onto the picture: temperature is about the statistics of enormous numbers of particles in motion, not about any single particle's individual behavior.

The Zeroth Law: What "Equilibrium" Really Means

Here's a deceptively simple everyday observation, elevated to the status of a fundamental law precisely because so much of thermodynamics depends on it: if you place a hot object in contact with a cold one, energy flows from hot to cold until both settle at the same temperature, and then nothing more happens — they've reached **thermal equilibrium**. If object A is in thermal equilibrium with object B, and B is separately in equilibrium with object C, then A and C are automatically in equilibrium with each other too, without ever needing to touch. This transitive property is called the **zeroth law of thermodynamics**, and it's what makes the entire concept of "temperature" logically well-defined in the first place: it guarantees that temperature is a single, consistent, comparable property, the same way "weighs more than" is a consistent comparison you can chain across any collection of objects.

Physically, equilibrium is a statement about balance: energy is still flowing back and forth at the microscopic level (individual fast molecules colliding with slow ones, exchanging energy every which way) but the *net* flow, averaged over all these countless collisions, is zero. Nothing observable is changing anymore, even though nothing has actually stopped moving.

Stellar Application: Local Thermodynamic Equilibrium

Here's an immediate puzzle if you try to apply this idea to a star. The Sun is most emphatically *not* in thermal equilibrium in any global sense — its core is around 1.5×10^7 K, its visible surface is around 5,800 K, and energy is continuously flowing outward from the hot core toward the cooler surface and out into the frigid vacuum of space, never stopping, never balancing out. If we insisted on true, global thermal equilibrium, the Sun simply wouldn't shine at all — and yet every layer, on the way from the Sun's core to its surface, still has a perfectly well-defined temperature. How can that be?

The resolution is a concept called **Local Thermodynamic Equilibrium (LTE)**. The idea is this: even though the star as a whole is wildly out of equilibrium (energy is steadily leaking out), if you zoom in on any sufficiently thin shell inside the star, the local conditions — the density, the rate of particle collisions, the rate of photon absorption and re-emission — are so overwhelmingly dominated by processes happening *within* that thin shell, compared to the comparatively slow trickle of energy leaking through it, that the local mix of particles and radiation settles into equilibrium

with each other, characterized by a single well-defined local temperature $T(r)$, even though that value of T smoothly changes from one shell to the next.

Think of it like a very long line of people passing buckets of water, hand to hand, to put out a fire far away. At any single person's position in the line, buckets are being exchanged constantly in both directions — receiving from one neighbor, passing to the next — and if you only watched one person's hands for a moment, it would look like a perfectly balanced, equilibrium exchange. It's only by looking at the *whole line* that you see there's a slow, steady net drift of water in one direction. LTE is exactly this idea, applied to matter and radiation inside a star: locally balanced, globally drifting.

LTE is what justifies treating each layer of a star as if it emits a blackbody spectrum at its local temperature (Section 3 of the Electromagnetic Theory & Quantum Physics handout), and it's what justifies using the Saha ionization equation and Boltzmann-distributed energy level populations *locally*, layer by layer, even for an object that is globally very far from equilibrium.

2 The Ideal Gas

Deriving Pressure From Scratch

Let's build up gas pressure from nothing but Newtonian mechanics — no formulas assumed, just molecules bouncing around in a box.

Picture a container of volume V holding N identical molecules, each of mass m , moving around randomly and colliding elastically with the walls (and with each other, though collisions between molecules don't change the total momentum delivered to the walls, so we can ignore them for this particular calculation). Consider one wall, a flat surface of area A perpendicular to the x -axis. When a molecule moving with x -velocity v_x hits this wall and bounces straight back elastically, its momentum changes by

$$\Delta p_x = m(-v_x) - m(v_x) = -2mv_x,$$

so the wall receives an impulse of $+2mv_x$ from that one collision.

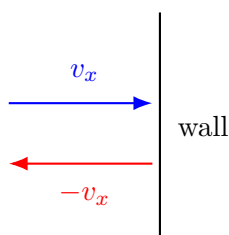


Figure 1: A molecule bouncing elastically off a wall reverses its x -velocity, delivering an impulse of $2mv_x$ to the wall.

How often does this happen? A molecule with x -velocity v_x , once it bounces off this wall, has to travel across the box (distance L , the box's length in the x -direction, so $V = AL$) and back before it can hit this same wall again — a round trip of $2L$, taking time $2L/v_x$. So this one molecule delivers an impulse of $2mv_x$ every $2L/v_x$ seconds, an average force contribution of

$$\frac{2mv_x}{2L/v_x} = \frac{mv_x^2}{L}.$$

Summing this over all N molecules in the box, and using $\overline{v_x^2}$ to denote the average of v_x^2 across all of them, the total force on the wall is $F = Nm\overline{v_x^2}/L$, and the pressure (force per unit area) is

$$P = \frac{F}{A} = \frac{Nm\overline{v_x^2}}{AL} = \frac{Nm\overline{v_x^2}}{V}.$$

Since the molecules move in completely random directions, there's no preferred axis, so $\overline{v_x^2} = \overline{v_y^2} = \overline{v_z^2}$, and since $v^2 = v_x^2 + v_y^2 + v_z^2$ for each molecule, we get $\overline{v^2} = 3\overline{v_x^2}$, i.e. $\overline{v_x^2} = \frac{1}{3}\overline{v^2}$. Substituting:

$$PV = \frac{1}{3}Nm\overline{v^2}.$$

Notice what we've done: derived a relationship between a macroscopic, easily measured quantity (pressure) and a microscopic, average property of countless individual molecules ($m\overline{v^2}$), using nothing but elastic collisions and bookkeeping. This is the heart of kinetic theory: macroscopic thermodynamic quantities are statistical summaries of microscopic mechanics.

Connecting to Temperature: the Ideal Gas Law

We can rewrite the boxed result as $PV = \frac{2}{3}N\left(\frac{1}{2}m\overline{v^2}\right)$, and the term in parentheses is just the average translational kinetic energy per molecule. If we now *define* temperature (for an ideal gas) via the equipartition relation

$$\frac{1}{2}m\overline{v^2} = \frac{3}{2}kT$$

(each of the three independent directions of motion carrying, on average, $\frac{1}{2}kT$ of energy, with $k \approx 1.38 \times 10^{-23}$ J/K the Boltzmann constant), substituting gives

$$PV = \frac{2}{3}N \cdot \frac{3}{2}kT \implies \boxed{PV = NkT}.$$

This is the **ideal gas law**, and notice that we haven't just quoted it — we've built it from the ground up, discovering along the way exactly what temperature *means* at the molecular level: it's a direct, linear measure of the average translational kinetic energy per particle. A gas at twice the absolute temperature genuinely has molecules carrying, on average, twice the kinetic energy — not some vaguer notion of "twice as hot."

A Quick Word on rms Speed

Rearranging the equipartition relation gives the root-mean-square speed of gas molecules directly:

$$v_{\text{rms}} = \sqrt{\overline{v^2}} = \sqrt{\frac{3kT}{m}}.$$

This is worth having on hand — it's what lets you compare, for instance, how fast hydrogen versus oxygen molecules move at the same temperature (lighter molecules move faster, for the same thermal energy), and it directly determines whether a planet's gravity is strong enough to retain a given gas over billions of years (compare v_{rms} to the escape velocity).

Stellar Application: Estimating the Sun's Core Temperature

Let's now use the ideal gas law to do something genuinely impressive: estimate the temperature at the center of the Sun, using nothing but order-of-magnitude reasoning and two equations.

The first equation is hydrostatic equilibrium (which you may recognize in spirit from the stellar structure discussion in the Celestial Mechanics handout): the outward push of gas pressure must support the star against its own gravity. In its exact differential form,

$$\frac{dP}{dr} = -\frac{GM_r\rho}{r^2},$$

but for an order-of-magnitude estimate, we can replace derivatives and radii with rough characteristic values: $dP/dr \sim -P_c/R$ (pressure drops from some central value P_c to roughly zero over the star's radius R), and $M_r \sim M$, $r \sim R$ on the right side. This gives

$$\frac{P_c}{R} \sim \frac{GM\rho}{R^2} \implies P_c \sim \frac{GM\rho}{R}.$$

The second equation is the ideal gas law itself, rewritten in terms of mass density $\rho = Nm/V$ (where m here is the average mass per gas particle) as $P = \rho kT/m$. Setting this equal to our estimate for P_c :

$$\frac{\rho kT_c}{m} \sim \frac{GM\rho}{R} \implies \boxed{T_c \sim \frac{GMm}{kR}}.$$

Notice the density ρ cancelled out entirely — a nice sign that this estimate is capturing something robust about the physics, not an artifact of a specific density profile. Plugging in solar values ($M_\odot \approx 2 \times 10^{30}$ kg, $R_\odot \approx 7 \times 10^8$ m, and taking m to be roughly the mass of a proton, since the Sun is mostly ionized hydrogen, $m_p \approx 1.67 \times 10^{-27}$ kg):

$$T_c \sim \frac{(6.67 \times 10^{-11})(2 \times 10^{30})(1.67 \times 10^{-27})}{(1.38 \times 10^{-23})(7 \times 10^8)} \approx 2 \times 10^7 \text{ K}.$$

This crude order-of-magnitude estimate lands remarkably close to the actual, carefully modeled core temperature of the Sun (about 1.5×10^7 K) — a genuinely satisfying result, given that we used nothing more sophisticated than "pressure balances gravity" and "pressure comes from an ideal gas," with no detailed density profile, no radiative transfer, none of the machinery a full stellar model actually requires. This is a good example of why order-of-magnitude physics is worth taking seriously in an exam setting: you very often don't need the full machinery to land within a factor of a few of the right answer.

3 Energy Transfer

Three Ways Energy Can Move

Energy generated in a star's core has exactly three physical mechanisms available to travel outward:

Conduction is energy passed directly from particle to particle via collisions, without the particles themselves making any net long-distance journey — like a message passed hand to hand down a tightly packed line of people, each person just nudging the next. In ordinary stellar material, conduction is usually hopelessly slow and can be ignored (it only becomes important in extremely

dense environments like white dwarfs, where free electrons conduct heat efficiently, much like in a metal).

Convection is energy carried by the bulk, physical movement of hot material itself — like a pot of boiling water, where hot blobs of fluid physically rise, carry their extra heat to the top, and cooler fluid sinks to take their place. This is a fast, efficient way to move energy, but only happens under specific conditions (Section 3.3).

Radiation is energy carried by photons themselves, streaming outward and being absorbed and re-emitted repeatedly along the way — like a very slow, very indirect game of “pass the parcel,” where a photon born deep inside the Sun doesn’t travel in a straight line to the surface at all, but instead scatters randomly, over and over, for an extraordinarily long time before it finally escapes. This last point deserves a full derivation, because the answer is far more dramatic than most people expect.

The Random Walk of a Photon

Deep inside the Sun, the gas is so dense and so ionized that a photon can only travel a very short distance — called the **mean free path**, ℓ — before it’s absorbed or scattered by an electron or ion, and re-emitted in a completely new, random direction. Because each step’s direction is independent of the last, the photon’s overall journey out of the star is a **random walk**, not a straight line.

Here’s the key mathematical fact about random walks, worth deriving carefully since it comes up constantly (not just for photons, but for diffusion of any kind): if each step has length ℓ in a random direction, what is the typical net *displacement* after N steps? Let each step be a vector $\vec{\ell}_i$ with $|\vec{\ell}_i| = \ell$. The squared displacement after N steps is

$$|\vec{R}|^2 = \left| \sum_{i=1}^N \vec{\ell}_i \right|^2 = \sum_i |\vec{\ell}_i|^2 + \sum_{i \neq j} \vec{\ell}_i \cdot \vec{\ell}_j.$$

The first sum is simply $N\ell^2$. The second sum, involving cross terms between *different* steps, averages to zero when you consider many such random walks — since each step’s direction is independent of every other, the dot product $\vec{\ell}_i \cdot \vec{\ell}_j$ is equally likely to be positive or negative, and these cancel out on average. So:

$$\boxed{\overline{R^2} = N\ell^2} \implies R_{\text{rms}} = \ell\sqrt{N}.$$

This is the single most important fact about random walks: the typical net distance traveled grows only as the *square root* of the number of steps, not linearly. To travel a net distance R therefore requires roughly

$$N \sim \left(\frac{R}{\ell} \right)^2$$

steps — and since it grows as the *square*, getting twice as far away requires four times as many random steps, not twice as many.

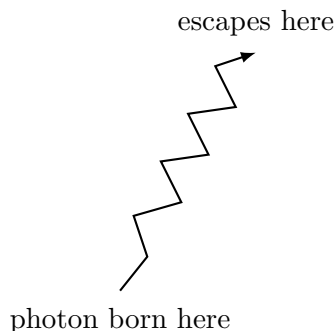


Figure 2: A photon's random walk out of the Sun. Each short segment (length ℓ , the mean free path) points in a completely new random direction — the net outward displacement after many such steps grows only as $\sqrt{N}$, not as N .

Worked Example: How Long Does Sunlight Take to Leave the Sun?

Now let's find the total time. If the photon takes $N \sim (R/\ell)^2$ steps, and each step (traveled at speed c) takes time ℓ/c , the total time is

$$t \sim N \cdot \frac{\ell}{c} = \left(\frac{R}{\ell}\right)^2 \cdot \frac{\ell}{c} = \frac{R^2}{\ell c}.$$

Using a mean free path inside the Sun of roughly $\ell \sim 1$ cm (a genuinely tiny distance, given how dense and opaque the solar interior is) and $R_{\odot} \approx 7 \times 10^{10}$ cm:

$$t \sim \frac{(7 \times 10^{10})^2}{(1)(3 \times 10^{10})} \approx 1.6 \times 10^{11} \text{ s} \approx 5,000 \text{ years}.$$

(Careful, more detailed calculations that account for the mean free path changing with depth give answers more like 10^4 – 10^5 years; the exact number is sensitive to assumptions, but the *scale* — thousands to tens of thousands of years, not seconds — is the genuinely important and surprising takeaway.)

Sit with how strange this is: the light warming your face right now left the Sun's surface just over eight minutes ago, but it was first *generated*, deep in the core, many thousands of years before that. Every photon reaching you today is a messenger from the Sun's ancient history, having spent millennia bouncing randomly through dense plasma before finally breaking free.

Convective Instability: The Schwarzschild Criterion

Radiation isn't always the dominant way energy escapes — sometimes it's simply too slow (exactly because of the random-walk bottleneck we just derived), and the star switches to the much faster mechanism of convection instead. When does this switch happen?

Picture a small parcel of gas nudged slightly upward from its surroundings. As it rises, it moves into a region of lower pressure and expands. If this expansion happens quickly compared to any heat exchange with its surroundings (a reasonable approximation deep inside a star), the parcel expands **adiabatically**, cooling as it does so at a specific, calculable rate — the adiabatic temperature gradient.

Now compare this to the star's own actual temperature gradient (how fast the background temperature falls off with radius, set by however the star transports energy at that depth). Two things can happen:

- If the risen parcel, after cooling adiabatically, is still *hotter* (and hence less dense) than its new surroundings, it keeps rising — buoyancy accelerates it further, exactly like a hot air balloon in cool air. This is a **convective instability**: any small upward nudge runs away.
- If the parcel cools down to match or fall below the surrounding temperature, it sinks back to where it started — a stable, non-convective (radiative) situation.

This comparison is the **Schwarzschild criterion**: a region is convectively unstable whenever the actual temperature gradient is *steeper* than the adiabatic gradient,

$$\left| \frac{dT}{dr} \right|_{\text{actual}} > \left| \frac{dT}{dr} \right|_{\text{adiabatic}}.$$

Physically, this tends to happen where the star's opacity is very high (radiation becomes an even slower way to move energy than the random-walk estimate above already suggests, forcing an even steeper temperature gradient to push the same energy flow through radiatively, which then trips the instability) or where energy generation is unusually concentrated.

Stellar Application: The Sun's Layered Structure

Put the pieces of this section together, and you get a genuine explanation for why the Sun has the layered structure it does. In the deep interior, densities and temperatures are high enough that radiative diffusion (however slow, per our random-walk estimate) is still fast enough to carry the core's energy output without tripping the Schwarzschild criterion — this inner region is called the **radiative zone**. Farther out, as temperature drops, the opacity of the increasingly neutral (rather than fully ionized) hydrogen and helium rises sharply, radiation becomes too inefficient a transport mechanism, the temperature gradient steepens past the adiabatic threshold, and the gas starts physically churning instead — the **convective zone**, extending from roughly 70% of the Sun's radius out to the visible surface.

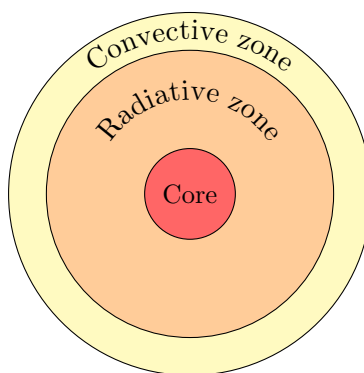


Figure 3: A schematic cross-section of the Sun. Energy generated by fusion in the core travels outward first via the slow random walk of radiative diffusion, then — once the Schwarzschild criterion is tripped — via bulk convective motion the rest of the way to the surface.

4 Putting It Together: A Star as a Thermodynamic System

It's worth stepping back and seeing how everything in this handout fits into the bigger picture of what actually determines a star's structure. A star, at any given radius r , is fully described by four coupled conditions, each one a direct descendant of a topic covered here or in the Celestial Mechanics handout:

1. **Hydrostatic equilibrium** (Celestial Mechanics handout, and used again above): $\frac{dP}{dr} = -\frac{GM_r\rho}{r^2}$ — gravity pulling in, pressure pushing out.
2. **Mass continuity** (Celestial Mechanics handout): $\frac{dM_r}{dr} = 4\pi r^2\rho$ — simple bookkeeping of how mass accumulates with radius.
3. **The equation of state** (this handout): $P = \frac{\rho kT}{m}$ — the ideal gas law, connecting pressure to density and temperature.
4. **Energy transport** (this handout): however energy actually moves outward — radiative diffusion or convection, whichever the Schwarzschild criterion favors at that depth — sets the local temperature gradient dT/dr .

These four equations, solved together (a task real stellar-structure codes handle numerically, but which we've now built enough intuition to estimate by hand, as in the core-temperature calculation above), are literally what determine everything about a star's structure: its central temperature and pressure, its radius, its luminosity, and ultimately its entire life story from birth to death. Thermodynamic equilibrium tells you a temperature is even meaningful to talk about at each layer; the ideal gas law tells you how that temperature relates to the pressure holding the star up; and energy transport tells you how the whole thing manages to shine steadily for billions of years without either collapsing or blowing itself apart.

5 Summary Sheet

Topic	Key Result
Zeroth law	Thermal equilibrium is transitive; makes temperature well-defined
LTE in stars	Local balance despite global energy outflow; justifies a local $T(r)$
Pressure, from kinetic theory	$PV = \frac{1}{3}Nm\overline{v^2}$
Equipartition	$\frac{1}{2}m\overline{v^2} = \frac{3}{2}kT$
Ideal gas law	$PV = NkT$
rms speed	$v_{\text{rms}} = \sqrt{3kT/m}$
Sun's core temperature, estimated	$T_c \sim GMm/(kR) \approx 2 \times 10^7 \text{ K}$
Random walk	$\overline{R^2} = N\ell^2$; $N \sim (R/\ell)^2$ steps to net distance R
Photon diffusion time	$t \sim R^2/(\ell c)$ — thousands of years for the Sun
Schwarzschild criterion	Convective if $ dT/dr _{\text{actual}} > dT/dr _{\text{adiabatic}}$